PG No. 34 $D_3(1)$ 32 (312 setting) [trigonal]

Table 1: Representation matrices

Irrep.						
A_1	$1:\begin{bmatrix}1\end{bmatrix}$	$3^+_{001}:\begin{bmatrix}1\end{bmatrix}$	$3^-_{001}:\begin{bmatrix}1\end{bmatrix}$	$2_{1-10}:\begin{bmatrix}1\end{bmatrix}$	$2_{120}:\begin{bmatrix}1\end{bmatrix}$	$2_{210}:\begin{bmatrix}1\end{bmatrix}$
A_2	$1:\begin{bmatrix}1\end{bmatrix}$	$3^+_{001}:\begin{bmatrix}1\end{bmatrix}$	$3^-_{001}:\begin{bmatrix}1\end{bmatrix}$	$2_{1-10}:\begin{bmatrix}-1\end{bmatrix}$	$2_{120}:\begin{bmatrix}-1\end{bmatrix}$	$2_{210}:\begin{bmatrix}-1\end{bmatrix}$
E	$1:\begin{bmatrix}1 & 0\\ 0 & 1\end{bmatrix}$	$3^+_{001}:\begin{bmatrix}-\frac{1}{2} & -\frac{\sqrt{3}}{2}\\ \frac{\sqrt{3}}{2} & -\frac{1}{2}\end{bmatrix}$	$3^-_{001}:\begin{bmatrix}-\frac{1}{2} & \frac{\sqrt{3}}{2}\\ -\frac{\sqrt{3}}{2} & -\frac{1}{2}\end{bmatrix}$	$2_{1-10}:\begin{bmatrix}\frac{1}{2} & -\frac{\sqrt{3}}{2}\\ -\frac{\sqrt{3}}{2} & -\frac{1}{2}\end{bmatrix}$	$2_{120}:\begin{bmatrix}-1 & 0\\ 0 & 1\end{bmatrix}$	$2_{210}:\begin{bmatrix}\frac{1}{2} & \frac{\sqrt{3}}{2}\\ \frac{\sqrt{3}}{2} & -\frac{1}{2}\end{bmatrix}$